

Functional Features

Project: Rising Water – AI-Based Flood Prediction System

1. Overview

This document describes the major functional features implemented in the Rising Water application.

Feature	Description
User Input	Accepts 20 environmental parameters through a web form.
Input Validation	Ensures all required values are entered correctly.
Data Preprocessing	Applies the saved scaler before prediction.
ML Prediction	Uses the trained XGBoost model to estimate flood probability.
Risk Classification	Displays Low, Medium, or High Risk.
Responsive UI	Provides a clean interface using HTML and CSS.
Fast Response	Returns prediction results within seconds.
Reusable Design	Supports future model updates without changing the UI.

User Workflow

1. Open the web application.
2. Enter all 20 flood-related parameters.
3. Click 'Predict Flood Risk'.
4. System validates and preprocesses inputs.
5. Machine learning model predicts flood probability.
6. Application displays probability and risk level.

Conclusion

The implemented functional features make the Rising Water system accurate, user-friendly, and suitable for flood risk assessment.